

Problem

Express the following signals in terms of singularity functions.

$$(a) \quad v(t) = \begin{cases} 0, & t < 0 \\ -5, & t > 0 \end{cases}$$

$$(b) \quad i(t) = \begin{cases} 0, & t < 1 \\ -10, & 1 < t < 3 \\ 10, & 3 < t < 5 \\ 0, & t > 5 \end{cases}$$

$$(c) \quad x(t) = \begin{cases} t-1, & 1 < t < 2 \\ 1, & 2 < t < 3 \\ 4-t, & 3 < t < 4 \\ 0, & \text{Otherwise} \end{cases}$$

$$(d) \quad y(t) = \begin{cases} 2, & t < 0 \\ -5, & 0 < t < 1 \\ 0, & t > 1 \end{cases}$$

Step-by-step solution

Step 1 of 4

(a)

$$v(t) = -5u(t)$$

Step 2 of 4

(b)

$$i(t) = -10[u(t-1) - u(t-3)] + 10[u(t-3) - u(t-5)]$$

$$i(t) = -10u(t-1) + 20u(t-3) - 10u(t-5)$$

Step 3 of 4

(c)

$$x(t) = (t-1)[u(t-1) - u(t-2)] + [u(t-2) - u(t-3)] \\ + (4-t)[u(t-3) - u(t-4)]$$

$$x(t) = (t-1)u(t-1) - (t-2)u(t-2) - (t-3)u(t-3) + (t-4)u(t-4)$$

$$x(t) = r(t-1) - r(t-2) - r(t-3) + r(t-4)$$

[Comments \(1\)](#)



Anonymous

how did you get this the theory is different

(d) $y(t) = 2u(-t) - 5[u(t) - u(t-1)]$

$$y(t) = 2u(-t) - 5u(t) + 5u(t-1)$$